Fawwaz Yaqzhan

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Summary

Final-year Computer Science student specializing in Computer Networks, DevOps, and Automation. Equipped with strong routing, switching, and security fundamentals through Cisco CCNA curriculum completion. Experienced in building backend systems and automating deployments using Python, Flask, and cloud databases. Adept at bridging software development with infrastructure operations.

Education

Padjadjaran University Sumedang Regency, West Java
Bachelor of Computer Science *Aug 2023 – Present*

Technical Skills

Languages: Python, SQL (MySQL, PostgreSQL, SQLite), JavaScript, HTML/CSS, Batch Scripting
Frameworks & Libraries: Next.js, Express, Flask, Flask-SQLAlchemy, pywebview, Chart.js, Jinja2
Developer Tools & Databases: Git, GitHub Actions, Nuitka, Supabase, Pinecone, Ollama, Midtrans, Packet Tracer, GNS3, Kali Linux, VS Code
AI Assist & Productivity: Google Gemini (AI-Assisted Development), Microsoft Excel

Work Experience

PT. EDS Manufacturing Indonesia (PEMI) - Yazaki Indonesia Group Tangerang Regency, Banten
Information Technology Intern *Jul 2026 – Aug 2026*

- **IT Inventory System**: Designed and developed a modular, full-stack web application using Flask, MySQL, and Vanilla JS/Chart.js, integrating bulk Excel imports and automated hardware lifecycle tracking for local corporate servers.
- Reconciled 2,500+ active PostgreSQL database records against procurement budgets using Excel to resolve cross-departmental financial discrepancies.
- Sanitized 50,000+ manufacturing inventory records on a live MySQL server, ensuring zero data corruption during manual log migrations.
- Supported IT operations and designed digital assets for Information Security Management System (ISMS) initiatives to promote anti-phishing awareness.

Organizational Experience

Padjadjaran University Sumedang Regency, West Java
Academic Department Staff *Feb 2024 – Dec 2024*

- Developed and produced peer-led educational video content addressing challenging computer science subjects based on student demand, enhancing remote learning accessibility on YouTube.
- Spearheaded introductory and mentoring sessions regarding the Student Creativity Program (PKM) to guide and inspire incoming freshmen toward academic research and innovation opportunities.
- Collaborated cross-functionally as a core committee member for IFEST UNPAD, assisting in the initial planning and synchronization for the National Seminar division.

Padjadjaran University Sumedang Regency, West Java
National Seminar Staff *Jul 2024 – Oct 2024*

- Served as a Liaison Officer (LO), acting as the primary bridge of communication and coordination between keynote speakers (industry experts) and the IFEST executive committee.
- Managed speaker hospitality, scheduling, and technical requirements during the event, ensuring a seamless presentation execution and high speaker satisfaction.

Projects

Smart Retail Network Architecture | Network Engineer May 2026 – Jun 2026
Cisco Packet Tracer, Cisco ISR 4331, Cisco Catalyst 3650, VLANs, Inter-VLAN Routing, ACLs, DHCP, WPA2-PSK

- Designed and simulated a secure network topology for a small-scale retail store in Cisco Packet Tracer, deploying a Cisco ISR 4331 edge router and Cisco Catalyst 3650 L3 core switch.
- Implemented network segmentation using VLANs and WPA2-PSK wireless access points to isolate operations for employees, IoT devices, and public guest Wi-Fi.

- Configured inter-VLAN routing on L3 switches and enforced Access Control Lists (ACLs) on the router to block guest Wi-Fi users from accessing the internal transaction server, database, and CCTV cameras.

Global Enterprise Network & Security Simulation | Network Engineer May 2026 – Jun 2026
GNS3, Cisco IOS, FortiGate NGFW, Kali Linux (DDoS Testing), VLANs, L3 Switching

- Simulated an enterprise network architecture for a global game company in GNS3, utilizing Cisco IOS Routers and L3 switches to connect development networks with central data and CDN servers.
- Integrated and configured a FortiGate Next-Generation Firewall (NGFW) to enforce strict security policies, protect core database servers, and manage traffic segmentation.
- Conducted security resilience assessments by simulating DDoS attacks using Kali Linux to test and verify the defense capabilities of the FortiGate firewall rules.

TanyaHukum (AI Legal Assistant) | Backend Developer Feb 2026 – Jun 2026
Node.js, Express, Next.js, PostgreSQL (Supabase), Pinecone, Midtrans, GitHub Actions, Gemini API

- Collaborated in a 5-member team to design and build the backend architecture, implementing RESTful APIs using Node.js and Express for user management and secure transactions.
- Integrated Gemini API with Pinecone Vector Database to construct a Retrieval-Augmented Generation (RAG) pipeline, delivering context-aware answers to legal queries.
- Designed secure authentication and Role-Based Access Control (RBAC) via Supabase and Google OAuth, separating permissions between regular users and administrators.
- Established a Continuous Integration (CI) pipeline using GitHub Actions to automate code validation and unit testing.

Chatbot JKKD (Course AI Assistant) | Fullstack Developer Jun 2026
Python, Flask, Ollama, Qwen 2.5, SQLite, HTML, CSS, Javascript, SQLite-Vector, Batch Files

- Developed a context-aware Question-Answering chatbot for the Computer Networks and Data Communication (JKKD) course using an Ollama-hosted LLM to assist students with queries.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline to fetch accurate answers from the course knowledge base, improving query resolution accuracy.
- Designed a lightweight Flask API and web interface to serve model responses locally, integrated with shell scripts for automated startup and deployment.

Capital Hub (Personal Financial Desktop App) | Fullstack Developer Aug 2026 – Present
Python, Flask, pywebview, Nuitka, SQLite, Chart.js, HTML5/CSS3, JavaScript

- Architected and compiled a standalone Windows desktop executable (.exe) financial management system using Flask bundled via Nuitka and pywebview, eliminating browser runtime dependencies.
- Engineered dynamic budgeting and cash flow tracking features paired with interactive Chart.js visualizations and an embedded SQLite database for local-first data privacy.
- Conducted UI/UX evaluation and application-level penetration testing to harden local data storage, sanitize inputs, and safeguard data integrity prior to public release.

Certifications & Languages

Certifications: CCNA Curriculum Completion (ITN, SRWE, ENSA) – Cisco Networking Academy | AWS Academy Graduate: Generative AI Foundations – AWS | English Language Test (Score: 553)

Languages: Indonesian (Native), English (Professional Working Proficiency)